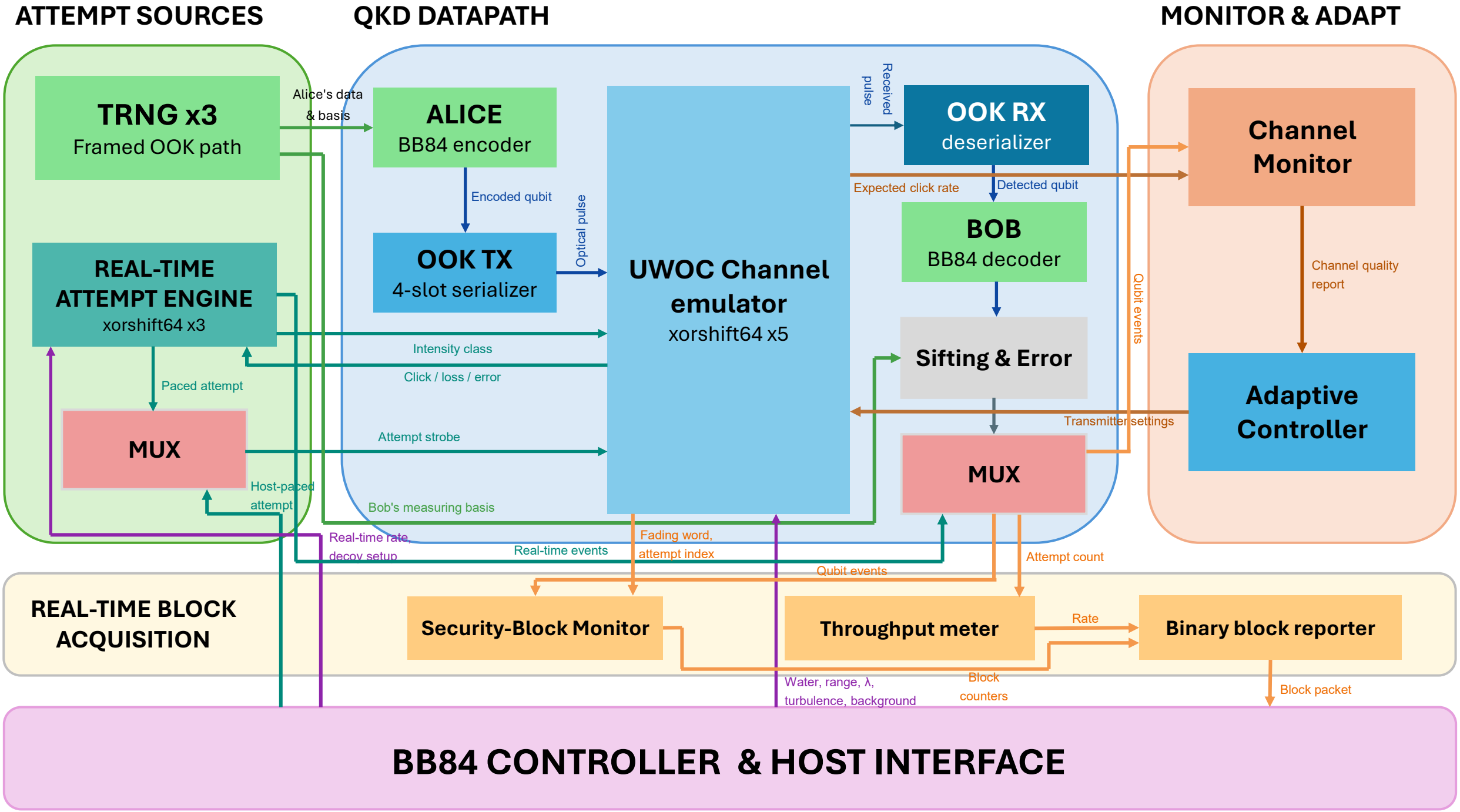

ATTEMPT SOURCES
QKD DATAPATH
MONITOR & ADAPT
TRNG x3
Framed OOK path
Alice's data & basis
ALICE
BB84 encoder
Encoded qubit
OOK TX
4-slot serializer
Optical pulse
UWOC Channel emulator
xorshift64 x5
Received pulse
OOK RX
deserializer
Expected click rate
Detected qubit
BOB
BB84 decoder
Sifting & Error
Qubit events
Channel Monitor
Channel quality report
Adaptive Controller
Transmitter settings
REAL-TIME ATTEMPT ENGINE
xorshift64 x3
Paced attempt
MUX
Intensity class
Click / loss / error
Attempt strobe
Host-paced attempt
Bob's measuring basis
Real-time rate, decoy setup
Real-time events
Fading word, attempt index
MUX
Attempt count
Qubit events
REAL-TIME BLOCK ACQUISITION
Security-Block Monitor
Throughput meter
Rate
Binary block reporter
Water, range, λ, turbulence, background
Block counters
Block packet
BB84 CONTROLLER & HOST INTERFACE